

Comparative Analysis of YOLOv8 Models for Thermal Image Person Detection

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Abstract—

This project presents a comparative analysis of YOLOv8n and YOLOv8s models for detecting persons in thermal images. The system was implemented using Python and the Ultralytics YOLOv8 framework. Three confidence thresholds (0.15, 0.25, and 0.50) were evaluated to analyze detection sensitivity and performance variation. Experimental results demonstrate that lower confidence thresholds increase detection counts, while higher thresholds significantly reduce detections. Although YOLOv8s achieved slightly higher detection results, YOLOv8n was selected due to its efficiency and comparable performance.

Keywords—

Thermal Imaging, Person Detection, YOLOv8, Deep Learning, Object Detection

I. INTRODUCTION

Thermal imaging technology has become increasingly important in surveillance systems, night monitoring, rescue missions, and security applications. Unlike traditional RGB cameras that rely on visible light, thermal cameras detect infrared radiation emitted by objects. This enables human detection even in darkness, fog, smoke, or low-visibility

conditions. However, detecting humans in thermal images presents several challenges. Thermal images lack color, texture, and facial details. Human shapes may appear unclear, and temperature similarities between objects can affect detection accuracy. Therefore, advanced deep learning-based object detection models are required to achieve reliable results. This project aims to implement and evaluate a thermal image person detection system using pre-trained YOLOv8 models. A comparison between two model variants was conducted, and different confidence thresholds were tested to analyze detection performance quantitatively.

II. PROBLEM DEFINITION

The objective of this project is to develop an automated system capable of detecting humans in thermal images using a pre-trained deep learning model.

The key challenges include:

- Limited visual detail in thermal images
- Reduced contrast and object boundaries
- Possible missed detections under strict confidence settings
- Sensitivity of results to parameter tuning

The system must process multiple thermal images and report the number of detected persons accurately.

III. RESEARCH PHASE – MODEL SELECTION AND COMPARISON

During the research phase, different object detection solutions were considered. Since training a model from scratch was not required, pre-trained models were selected.

The YOLO (You Only Look Once) family of models was chosen due to:

- Real-time detection capability
- High performance in object detection tasks
- Availability of lightweight and scalable versions
- Easy implementation using the Ultralytics Python library

Two variants were selected for comparison:

- YOLOv8n (Nano version – lightweight and fast)
- YOLOv8s (Small version – larger and more powerful)

Feature	YOLOv8n	YOLOv8s
Model Size	Very small	Small
Speed	Very Fast	Fast
Computational Cost	Low	Medium
Number of Parameters	Fewer	More
Expected Accuracy	Moderate	Higher

YOLOv8n is optimized for speed and low computational resources.

YOLOv8s contains more parameters, potentially improving detection accuracy but requiring more processing power.

The goal was to determine whether the larger model significantly improves detection performance in thermal images.

IV. METHODOLOGY (Implementation Phase)

The implementation was performed using:

- Python programming language
- Ultralytics YOLOv8 library
- OpenCV for image processing

The workflow of the system:

- Load a pre-trained YOLOv8 model
- Read thermal images from a folder
- Detect only the “person” class
- Apply different confidence thresholds
- Draw bounding boxes
- Save detection results
- Record total detections

Three confidence thresholds were tested:

- 0.15
- 0.25
- 0.50

Lower confidence increases sensitivity (detects more objects).

Higher confidence increases strictness (reduces weak detections).

The dataset consisted of 4 thermal images containing approximately 15 persons (estimated manually).

V. EXPERIMENTAL RESULTS

YOLOv8n Results

Confidence	Total Detections
0.15	12
0.25	10
0.50	5

YOLOv8s Results

Confidence	Total Detections
0.15	13
0.25	12
0.50	6

Ground Truth Estimate

The approximate true number of persons in all images combined: 15 persons.

VI. ANALYSIS AND DISCUSSION

The results show a clear relationship between confidence threshold and detection count.

At confidence 0.15:

- Most persons were detected.
- Detection count was closest to the estimated ground truth (15).

At confidence 0.25:

- Some persons were missed.
- Results became more conservative.

At confidence 0.50:

- Detection dropped significantly.
- Many valid persons were not detected.

Comparison between YOLOv8n and YOLOv8s showed:

- YOLOv8s detected slightly more persons at lower confidence.
- However, the improvement was not significant.
- The larger model did not dramatically outperform the nano version.

This indicates that model size alone does not guarantee better performance in thermal imaging conditions without fine-tuning.

VII. FINAL MODEL SELECTION

Although YOLOv8s showed slightly higher detection numbers, the difference was small compared to YOLOv8n.

Considering:

- Similar detection performance
- Lower computational cost
- Faster processing speed
- Better efficiency

YOLOv8n was selected as the final model for this project.

This decision reflects a balance between performance and practicality.

VIII. CONCLUSION

This project successfully implemented a thermal image person detection system using pre-trained YOLOv8 models.

Two model variants (YOLOv8n and YOLOv8s) were compared under different confidence thresholds. The experimental evaluation demonstrated that lower confidence thresholds increase detection sensitivity, while higher thresholds reduce detections significantly.

Although YOLOv8s slightly improved detection results, the performance gain was not substantial. Therefore, YOLOv8n was selected as the final model due to its efficiency and comparable performance.

Future improvements may include:

- Fine-tuning the model on thermal-specific datasets
- Increasing dataset size
- Measuring precision and recall formally

- Testing additional object detection architectures

Overall, the project demonstrates that pre-trained YOLOv8 models can be effectively adapted for thermal person detection tasks.

IX. REFERENCES

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